Kristina Isabella E. Dee

ked2188@columbia.edu · (347) 827-9996 · linkedin.com/in/kristinaidee · kristinadee.github.io

Mechanical engineer building image-based, patient-specific finite element models and 3D visualizations of soft tissue for women's health, with hands-on CAD, prototyping, and test experience for carrying models into medical devices.

EDUCATION

Columbia University, B.S. Mechanical Engineering | Major GPA 3.5 Expected May 2027

Relevant coursework: Mechanics of Solids, Fluid Mechanics, Heat Transfer, Dynamics & Vibrations, Computer Graphics & Design, Mechatronics & Embedded Systems, Materials & Processes in Manufacturing

In progress (Fall 2026): Engineering Design (senior design), Machine Design, Continuous Control Systems

RESEARCH EXPERIENCE

Computational Biomechanics Research Lab (Prof. Vijay Vedula) | *Patient-Specific Uterine Mechanics* Apr 2025 – Present

- Scaled an image-based uterine and cervical mechanics pipeline from a single-patient proof of concept to a **5-patient cohort**, building **4 new patient models** and running **10 nonlinear FE simulations** (FEBio and svMultiPhysics; **580,749 nodes**).
- Benchmarked spatially varying **Robin boundary conditions** (adapted from cardiac mechanics) against a contact-based FEBio reference: **3.6 ± 2.2 mm** pooled mean nodal displacement difference; BC parameters tuned on one patient **transferred to three others without re-tuning** (means within 0.4 mm).
- Built the imaging-to-simulation workflow: parametric **SolidWorks** geometry from ultrasound/MRI measurements, continuous-node **HEX8/TET4 meshes** in HyperMesh, **Laplace–Dirichlet rule-based fiber fields**, and the **GOH anisotropic hyperelastic** model.
- Ran MPI-parallel jobs via **SLURM** on Columbia Ginsburg and ACCESS SDSC Expanse, cutting time per case **40%**; resolved fillet stress concentrations, node continuity, and mesh-conversion bottlenecks.
- Produced 3D, sagittal, and coronal displacement-difference maps and fiber-field renderings in **ParaView**.

Building Energy Research Lab | *Reinforcement Learning for HVAC Control* Sep 2024 – Sep 2025

- Trained **Soft Actor-Critic** policies (Python/PyTorch) on FMU-based physics simulation (Dymola/OpenModelica), reaching **±0.5 °C** setpoint accuracy within 50–75 episodes and outperforming PPO and DQN baselines.
- Validated **sim-to-real transfer** on a 1,500–2,000 sq ft building testbed; swept 5 hyperparameter families.

PUBLICATIONS & PRESENTATIONS

- Mergler, O., **Dee, K.**, et al. *A Method for Image-Based Modeling of Uterine Passive Mechanics During Late Pregnancy.* (In Revision for *Annals of Biomedical Engineering*). Preprint: bioRxiv 10.64898/2026.07.10.737823
- **Dee, K.**, Mergler, O., Myers, K., Vedula, V. *Multi-Patient Evaluation of Computational Models of Uterine Mechanics.* Poster, Columbia University Undergraduate Research Symposium, New York, NY, Oct. 23, 2026.

DESIGN PROJECTS

Senior Design: Accessible Menstrual Disc Inserter | *Team of 5* Sep 2026 – Present

- Designing an assistive inserter that lets people with limited hand dexterity or mobility and heavy flow place a menstrual disc independently; translating user needs into design requirements and early concepts in **SolidWorks**.

Humanity Without Borders | *Founder / Director* Jul 2020 – Aug 2023

- Built a **CNN for satellite-image damage classification (95% accuracy)** on xBD and multilingual NLP needs-assessment pipelines for disaster-response agencies.

TECHNICAL EMPLOYMENT

Nanov Display Inc. | *Technical Intern* Jan 2025 – May 2025

- Designed weather-resistant kiosks and transit LCD/LED signage in SolidWorks/AutoCAD; released drawings and **3D renderings**.
- Set up new LED and LCD displays end to end: all electrical and PCB-level wiring, embedded system configuration (Windows, Ubuntu), and **Factory Acceptance Testing** with compliance documentation.
- As the only engineer handling them at the small Brooklyn office, led **technical-support meetings with suppliers** and worked with them to reduce BOM cost; hosted **visiting clients** on tours of the warehouse equipment and kiosks.

TECHNICAL SKILLS

Simulation & Biomechanics: nonlinear FEA (FEBio, svMultiPhysics), soft-tissue mechanics, GOH hyperelasticity, fiber modeling
Geometry, Meshing & Visualization: ultrasound/MRI-based geometry, SolidWorks, HyperMesh (HEX8/TET4), ParaView
Computing: Python (NumPy, SciPy, PyTorch), MATLAB, C++, HPC/MPI (SLURM), Linux, reinforcement learning (SAC, TD3), CNNs, NLP
Design & Prototyping: AutoCAD, drawings, BOM, acceptance testing, CNC milling (Tormach), G-code/CAM, mill/lathe, metrology